

John Marsden

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[LinkedIn](#) | [GitHub \(Professional\)](#) | [GitHub \(Personal\)](#)

EDUCATION

North Carolina State University

M.S. – Computer Science

Graduation Date: 04/2025

Abilene Christian University

B.S. – Math and Computer Science

Graduation Date: 04/2020

SKILLS

Databases: MySQL, Oracle Database, Postgres

Dev Ops: Bash, Caddy, DigitalOcean, Docker, Github Actions, Heroku, Linux, Nginx, Redis

Frameworks: ASP.NET, Astro, AsyncAPI, Django, Flask, GraphQL, jQuery, Node.js, Puppeteer, Rails, Ruby on Rails, Selenium, Tornado, Vue

Programming Languages: C, C++, C#, CSS, F#, Haskell, HTML, Java, JavaScript, Kotlin, Lua, PHP, Prolog, Python, Ruby, Rust, SQL, Typescript

EXPERIENCE

Teaching Assistant | [NCSU](#)

07/2020 – 11/2024

- Helped 300+ students navigate the complexities of multi-project **Java** codebases
- Graded 400+ student algorithms using common data structures, including stacks, graphs, etc.

Instructor - Software Development Fundamentals | [NCSU](#)

05/2022 – 07/2022

- Taught the software development process and **test-driven development** to 20 students
- Demonstrated idiomatic generic object-oriented **Java** code to students during 5+ live coding sessions

Instructor - Introduction to Computing | [NCSU](#)

12/2021 – 04/2022

- Taught the fundamentals of **Java** to 40 students
- Built two autograders using **JUnit 5** to autograde student code submissions

Software Development Intern | [Secmation](#)

04/2021 – 07/2021

- Planned and implemented 20% of a backend for a programming environment with **Python**
- Outlined an API for the programming environment backend using **AsyncAPI**
- Containerized the programming environment backend using **Docker** for scalability and ease of use

Report Specialist | [ACU](#)

09/2017 – 04/2020

- Crafted SQL queries to aggregate raw data about alumni interaction into usable information for reporting
- Built Cognos reports to present data in a user-friendly format for relevant stakeholders

Software Development Intern | [Faithlife](#)

04/2019 – 07/2019

- Converted the **OAuth** application management portal from being user specific to a group-based system
- Integrated a **HaveIBeenPwned** check using the **k-anonymity** API

PROJECTS

Control Flow Graph Compiler

- Implemented a custom intermediate language representation with the necessary semantics for converting programs to control flow graphs (CFG)
- Created a compiler in **Rust** to convert the custom intermediate language to a statement level control flow graph
- Implemented a parallelized **TCP** server on top of the compiler to handle over two hundred thousand compilations necessary for training a **machine-learning** model

Machine Learning on Intermediate Languages

- Trained a code embedding machine learning model to convert control-flow graphs (CFG) to vectors
- Reimplemented the deepwalk learning algorithm for distributed learning
- Constructed an attention network for using CFG embeddings to predict student performance

General Purpose Discord Bot

- Used **dependency injection** to maintain separation of concerns and allow for easy testing
- Implemented an **HTML** to Discord Markdown **transpiler** to convert **RSS** feed posts to Discord embeds
- Built an event announcement system that supported **timezones** and role-based announcements
- Wrote an OMDB API client using **System.Text.Json** and **RestClient** to allow users to plan movie nights with my designed command flow

Natural Language Processing Wrapper Library

- Designed a wrapper library around the Microsoft.Recognizers.Text library adding static type support
- Created an extensible structure for supporting future recognizers
- Authored a documented library API for ease of future extension and contribution